WHAT?

The presence of Dimethyl Sulfide (DMS) or Dimethyl Disulfide (DMDS), or both, in its atmosphere. These chemical compounds are uniquely produced on Earth by phytoplankton in the oceans

HOW?

As the planet transits in front of its host star, the starlight goes through its atmosphere. The James Webb Space Telescope, which is in its own solar orbit, analysed this light through **transit spectroscopy** for clues about K2-18b's atmosphere

WHERE?

K2-18b: A planet 124 light years away, orbiting a red dwarf star. It is about two and a half times the size of earth, and rich in hydrogen—a prerequisite for life as we know it

HOW TRUE?

The results are now at the level of three sigma: a probability of 99.7 per cent. Science demands five sigma, or 99.99994 per cent

WHAT NEXT?

Stronger evidence is required, especially to rule out an 'abiotic' (non-life) source for DMS on this exoplanet

K2-18b
2.6 times the radius of Earth

124
LIGHT YEARS



